CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (ME)

May 2018

ME-309 Design of Machine Elements-I

Date: 01.05.2018, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 (a) Give Answer of Following MCQ's. **[07]**

1. Which one of the following loading is considering for the axle design?
- a. Only bending moment
- b. Only twisting moment
- c. Combined bending and twisting
- d. Combined bending, twisting and axial thrust

2. The type of key is used when the gear is required to slide on the shaft is
- a. Sunk
- b. Feather
- c. Woodruff
- d. Kennedy

3. In designing of the sleeve coupling, outer diameter of the sleeve is taken as
- a. d + 17 mm
- b. 2d + 13 mm
- c. 2d + 20 mm
- d. 3.5 d

4. In forged components,
- a. Fiber lines are in predetermined way
- b. Fiber lines of rolled stroke are broken
- c. There are no fiber lines
- d. Fiber lines are scattered

5. In case of clamp coupling, power is transmitted by means of
- a. Frictional force
- b. Shear resistance
- c. Crushing resistance
- d. None of the above

6. Cold working
- e. Increases fatigue strength
- f. Decreases fatigue strength
- g. Has no influence on fatigue strength
- h. None of the above

7. Series factor for R20 series is,
- a. $\sqrt[10]{20}$
- b. $\sqrt{20}$
- c. $\sqrt[20]{10}$
- d. $\sqrt[3]{20}$

Q – 1 (b) A transmission shaft supporting a spur gears B and pulley D is shown in **Fig 1**. The shaft is mounted on two bearings A and C. the diameter of pulley and the pitch circle diameter of gear are 450 and 300 mm respectively. The pulley transmits 20 kW power at 500 rpm to the gear. P_1 and P_2 are belt tensions in the tight and loose sides, while P_t and P_r ate tangential and radial components of gear tooth force. **[09]**
Assume, $P_1 = 3P_2$ and $P_r = P_t \tan 20^{o.}$ The gear and pulley are keyed to the shaft. The material of the shaft is steel 50C4 having $S_{ut} = 700$ N/mm^2 and $S_{yt} = 460$ N/mm$^{2.}$ The factors K_b and K_t of ASME code are 1.5 each. Determine the shaft diameter using ASME code.

OR

(b) 1. It is required to design a square key for fixing a gear on a shaft of 25 mm diameter. The shaft is transmitting 15 kW power at 720 r.p.m. to the gear. The key is made of **[04]**

steel 50C4 having S_{yt} = 460 N/mm^2 and the factor of safety is 3. For key material, the yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key.

2. Design a muff coupling to connect two steel shafts transmitting 25 kW power at 360 r.p.m. the shaft is made of plain carbon steel having S_{yt} = S_{yc} = 400 N/mm$^{2.}$ The sleeve is made of grey cast iron with S_{ut} = 200 N/mm$^{2.}$ The factor of safety for the shaft is 4 and for sleeve is 6 based on S_{ut}. **[05]**

Q – 2 Answer the following Questions.

1. A forged steel bar, 50 mm in diameter, is subjected to a reversed bending stress of 250 N/mm^2. The bar is made of steel 40C8 (S_{ut} - 600 N/mm^2). Calculate the life of the bar for a reliability of 90%. Assume K_a = 0.44. **[05]**

2. Define the following terms: **[02]**

- **a.** Stress concentration
- **b.** Fatigue life

3. Explain the general principles for the design of casting with neat sketches. **[06]**

Q – 3 Explain all the types of cyclic stresses with neat sketch. **[06]**

OR

Q – 3 Explain the modified goodman diagram with neat sketch. **[06]**

SECTION – II

Q - 4 Give Answer of Following MCQ's. **[07]**

1. The coefficient of friction between the belt and pulley depends upon the

a. Material of belt and pulley b. Slip of belt
c. Speed of belt d. All of these

2. For a chain drive with variation of speed less than 1% the maximum number of teeth on smaller sprocket should be

a. 15 b. 17
c. 20 d. 24

3. Multiple threaded screws are used for

a. High travelling speed b. High mechanical advantage
c. High load capacity d. None of the above

4. In case of thick cylinders, the tangential stress across the thickness of cylinder is

a. maximum at the outer surface and minimum at the inner surface
b. maximum at the inner surface and minimum at the outer surface
c. maximum at the inner surface and zero at the outer surface
d. maximum at the outer surface and zero at the inner surface

5. The design of the pressure vessel is based on

a. longitudinal stress b. hoop stress
c. longitudinal and hoop stress d. None of the above

6. The clutch is located between the transmission and the

a. Engine b. Rear axle
c. Propeller shaft d. Differential

7. When the frictional force helps to apply the brake, then the brake is said to be self-energizing brake.

a. Correct b. Incorrect

Q – 5 Answer the following Questions.

1. What are the methods of pre-stressing the cylinder? **[03]**

2. A steel tube 240 mm external diameter is shrunk on another steel tube of 80 mm internal diameter. After shrinking, the diameter at the junction is 160 mm. Before shrinking, the difference of diameters at the junction was 0.08 mm. If the Young's modulus for steel is 200 Gpa, **[06]**

Find: 1. Tangential stress at the outer surface of the inner tube, 2. Tangential stress at the inner surface of the outer tube, and 3. Radial stress at the junction

3. Illustrate all the condition of short shoe block brake with neat sketch. **[06]**

OR

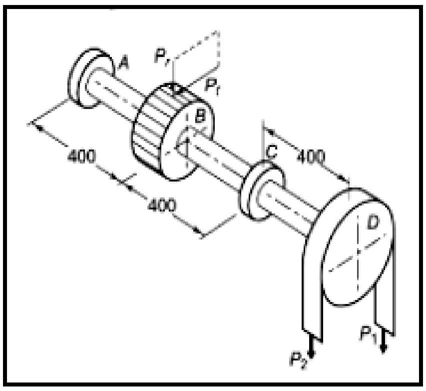
1. An air receiver consisting of a cylinder closed by hemispherical ends is shown in **Fig. 2.** It has a storage capacity of 0.25 m3 and an operating internal pressure of 5 MPa. It is made of plain carbon steel 10C4 (σ_{ut} = 340 N/mm^2) and the factor of safety is 4. Neglecting the effect of welded joints, determine the dimensions of the receiver. **[03]**

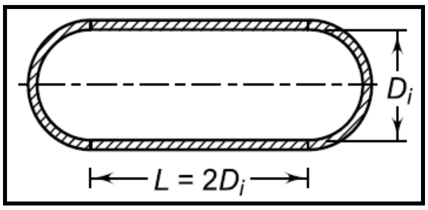
2. A belt pulley, made of grey cast iron FG 150, has four arms of elliptical cross section, in which the major axis is twice of the minor axis. The tensions on tight and slack sides of the belt are 150 and 250 N respectively. The mean diameter of the pulley is 300 mm; while the hub diameter 60 mm. assume that half number of arms transmit torque at any time. The factor of safety is 5. Determine the dimensions of the cross section of the pulley arm near the hub. **[06]**

3. Write down the components of chain with neat sketch. Why the sprockets has odd number of teeth and the chain has even number of links? **[06]**

Q - 6 (a) A sluice gate weighing 18 kN is raised and lowered by means of square threaded screws, as shown in **Fig. 3**. The frictional resistance induced by water pressure against the gate when it is in its lowest position is 4000 N. The outside diameter of the screw is 60 mm and pitch is 10 mm. The outside and inside diameter of washer is 150 mm and 50 mm respectively. The coefficient of friction between the screw and nut is 0.1 and for the washer and seat is 0.12. Find: 1. The maximum force to be exerted at the ends of the lever raising and lowering the gate, 2. Efficiency of the arrangement, and 3. Number of threads and height of nut, for an allowable bearing pressure of 7 N/mm^2 **[08]**

(b) Derive an equation of torque transmitting capacity of single plate clutch for uniform wear condition. **[05]**





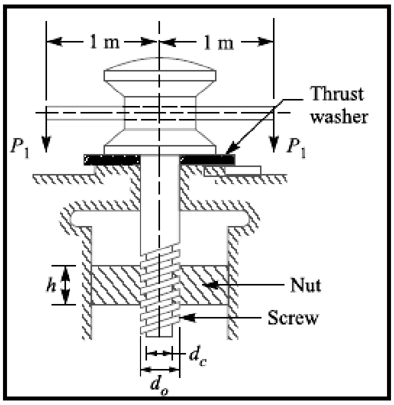


Fig. 1 **Fig.2** **Fig.3**